NALA-DNA-Med aPP

Quellen- und Datenblatt für medizinisches Personal, Ausschüsse und Codex-Implementierung

Version: 0.1 Konzept / MVP-Quellenpaket · Stand: 27. Mai 2026 · Repo: https://github.com/Master-MD/NALA-DNA-Med-

Dieses Dokument sammelt die verwendeten Datenquellen, Original-Kernaussagen in kurzer Zusammenfassung und die jeweilige Relevanz für das NALA-DNA-Med Konzept.

Wichtiger medizinischer Hinweis

NALA-DNA-Med ist in diesem Konzept ein lokales Forschungs-, Dokumentations- und Hypothesen-System. Es ist nicht validiert für Diagnose, Therapieentscheidung, Medikamentenverschreibung oder klinische Entscheidungsfindung. Alle BioModel-Ausgaben müssen als Research Support / Demo / Hypothesis Generation gekennzeichnet werden.

Erstellt für: Michael Dreher / NALA-cORe · Ziel: vorzeigbare, quellenbasierte Dokumentation für Arztgespräch und Projektfreigabe.

1. Executive Summary

Die Quellenlage stützt ein sinnvolles MVP: Eine lokale, privacy-first Plattform mit Multi-User-Zugriff, strikter Tenant-Isolation, RAG-Vault, BioModel-Service als Stub und späterer MAMMAL/RDKit-Integration. Die Kernentscheidung bleibt: UI/Local Lite direkt auf dem MacBook Pro, robuste Core-Dienste über OrbStack/Docker Compose.

Die wichtigste technische Leitplanke ist Mandantenfähigkeit: gemeinsame Modelle/GPU-Ressourcen ja; private Daten, Dokumente, Vektoren, Jobs und Logs strikt pro Tenant/User/Projekt getrennt.

Empfohlener MVP-Schnitt

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Local Lite Mode: UI + API + Demo-Analyse ohne Docker.
OrbStack Core Mode: Postgres + Qdrant + MinIO + Redis + API + BioModel API + UI.
BioModel v0.1: Stub-Endpunkte, keine grossen Modelldownloads, keine medizinischen Claims.
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Risiko-Einschätzung

MVP mit User/Projektstruktur und isoliertem RAG-Design: 85% success / 10% unknown / 5% fail. Echte MAMMAL-Integration lokal: 65% success / 25% unknown / 10% fail, weil Datenformate, Dependencies und Apple-MPS-Kompatibilität geprüft werden müssen.

2. Architektur-Entscheid aus den Quellen

Shared compute, isolated data

The model backend, GPU, and common public model weights can be shared; private user uploads, chats, vectors and audit logs must be separated by tenant_id/user_id/project_id.

Local Lite Mode

Runs on macOS without Docker. It should provide UI, demo analysis, project vault and health checks. It should not attempt heavy model downloads in v0.1.

OrbStack Core Mode

Uses Docker Compose for Postgres, Qdrant, MinIO, Redis, API, BioModel API and UI. This is cleaner for reproducible multi-user operation.

BioModel as plugin

MAMMAL and RDKit should be integrated later through a dedicated Python/FastAPI service. The v0.1 app should contain only demo endpoints and disclaimers.

Medical safety

Outputs must be labelled as research support/hypothesis generation, not diagnosis, treatment or validated medical advice.

3. Quellen mit Kurz-Zusammenfassung

Biomedical Foundation Model / MAMMAL

1. GitHub: BiomedSciAI/biomed-multi-alignment

https://github.com/BiomedSciAI/biomed-multi-alignment

Original-Kernaussage / Zusammenfassung: Repository describes ibm/biomed.omics.bl.sm.ma-ted-458m as a biomedical foundation model trained on over 2 billion biological samples across proteins, small molecules and single-cell gene data. It positions the model for tasks across the drug-discovery pipeline and provides code/weights references under Apache-2.0.

Relevanz für NALA-DNA-Med: Primary technical source for later MAMMAL integration. For v0.1 the app should only include a stub service and integration notes, not auto-download large model weights.

2. Hugging Face: ibm-research/biomed.omics.bl.sm.ma-ted-458m

https://huggingface.co/ibm-research/biomed.omics.bl.sm.ma-ted-458m

Original-Kernaussage / Zusammenfassung: Model card states that the model is a biomedical foundation model trained on over 2 billion biological samples across modalities including proteins, small molecules and single-cell gene data. It is designed for robust performance across biomedical and drug-discovery tasks.

Relevanz für NALA-DNA-Med: Reference for model download location, model size family (~0.5B), intended input domains, and future plugin target.

3. arXiv: MAMMAL - Molecular Aligned Multi-Modal Architecture and Language

https://arxiv.org/abs/2410.22367

Original-Kernaussage / Zusammenfassung: Paper presents MAMMAL as a multi-modal biomedical architecture aligning molecular, protein and omics data. It states that model code and pretrained weights are publicly available through GitHub and Hugging Face.

Relevanz für NALA-DNA-Med: Scientific background for the BioLab concept. Use as citation for why molecule/protein/omics workflows belong in the app, while still labelling outputs as research-support only.

Biomedical Foundation Models / IBM

1. IBM Research: Biomedical Foundation Models

https://research.ibm.com/projects/biomedical-foundation-models

Original-Kernaussage / Zusammenfassung: IBM Research describes biomedical foundation model technologies that use multimodal biomedical data such as drug-like small molecules, proteins and single-cell RNA sequence data, with research expertise spanning chemistry, AI, biology and biomedical informatics.

Relevanz für NALA-DNA-Med: High-level institutional context for committees/doctors: this approach is an active research direction, not a random hobby stack.

Multitenancy / Data Isolation

1. Qdrant Documentation: Multitenancy

https://qdrant.tech/documentation/manage-data/multitenancy/

Original-Kernaussage / Zusammenfassung: Qdrant recommends, in many cases, using a single collection per embedding model with payload-based partitioning for tenants and use cases. The documentation also discusses tiered multitenancy and per-tenant isolation/performance considerations.

Relevanz für NALA-DNA-Med: Core source for RAGVault design: every vector payload must include tenant_id, user_id, project_id and document_id, and every retrieval query must enforce filters.

2. Qdrant Article: How to Implement Multitenancy and Custom Sharding

https://qdrant.tech/articles/multitenancy/

Original-Kernaussage / Zusammenfassung: Qdrant article explains that vectors can share a collection while being separated by tenant-specific payload metadata. Tenant payloads enable filtering and large-scale multitenant setups.

Relevanz für NALA-DNA-Med: Supports the TenantShield rule: shared infrastructure is OK, but every stored vector/document must carry tenant metadata.

Multi-User Frontend / RBAC

1. Open WebUI Documentation: Features / Authentication & Access

https://docs.openwebui.com/features/

Original-Kernaussage / Zusammenfassung: Open WebUI documentation says the platform is multi-user from day one, with roles, groups, per-model access and identity-provider integration.

Relevanz für NALA-DNA-Med: Good MVP option for a ready-made multi-user chat UI before building a fully custom NALA Black Edition interface.

2. Open WebUI Documentation: Role-Based Access Control

https://docs.openwebui.com/features/authentication-access/rbac/

Original-Kernaussage / Zusammenfassung: RBAC guide covers roles, permissions, groups and security/privacy configuration. It explicitly separates role management from granular permission toggles.

Relevanz für NALA-DNA-Med: Reference for roles such as Admin, User, Pending, and for designing doctors/researcher/viewer/admin workflows.

3. Open WebUI Documentation: Roles

https://docs.openwebui.com/features/authentication-access/rbac/roles/

Original-Kernaussage / Zusammenfassung: Open WebUI defines primary system roles: Admin as superuser with full control, User as standard RBAC-governed account, and Pending as restricted until approved.

Relevanz für NALA-DNA-Med: Useful for CAVEMAN-level role mapping: Owner/Admin/Doctor/Researcher/Viewer/Guest/Pending.

LLM Serving / Model Backend

1. vLLM Documentation: OpenAI-Compatible Server

https://docs.vllm.ai/en/stable/serving/openai_compatible_server/

Original-Kernaussage / Zusammenfassung: vLLM provides an HTTP server implementing OpenAI-compatible Completions and Chat APIs so models can be served and accessed via standard HTTP clients.

Relevanz für NALA-DNA-Med: Supports NALA-cORe backend strategy: serve local LLMs behind an OpenAI-compatible interface for UI/agent compatibility.

2. vLLM Documentation: Quickstart

https://docs.vllm.ai/en/stable/getting_started/quickstart/

Original-Kernaussage / Zusammenfassung: Quickstart explains vLLM can be deployed as a server implementing the OpenAI API protocol and used as a drop-in replacement for OpenAI API applications.

Relevanz für NALA-DNA-Med: Good source for why NALA can support interchangeable local/remote LLM backends with the same app interface.

macOS Runtime / Container Mode

1. OrbStack Docs: Docker Containers

https://docs.orbstack.dev/docker/

Original-Kernaussage / Zusammenfassung: OrbStack includes a Docker engine for running containers. Networking, port forwarding, bind mounts and volumes work with Docker-compatible workflows.

Relevanz für NALA-DNA-Med: Main source for choosing OrbStack Core Mode for Qdrant/Postgres/MinIO/Redis/BioModel workers on Mac.

2. OrbStack Quick Start

https://orbstack.dev/docs/quick-start

Original-Kernaussage / Zusammenfassung: Quick Start describes setting up Docker containers and full Linux machines on Mac, with Homebrew install option and simple docker run example.

Relevanz für NALA-DNA-Med: CAVEMAN install source: install OrbStack, run Docker Compose, then start NALA Core Mode.

3. OrbStack Native Files / Container, Image, Volume Files

https://docs.orbstack.dev/features/native-files

Original-Kernaussage / Zusammenfassung: OrbStack exposes container, image, volume and machine files natively to macOS through Finder and CLI-friendly paths.

Relevanz für NALA-DNA-Med: Important UX benefit for Mike/NALA: users can inspect container files without deep Docker kung-fu.

Apple Silicon / Local Inference

1. Apple Developer: Accelerated PyTorch Training on Mac

https://developer.apple.com/metal/pytorch/

Original-Kernaussage / Zusammenfassung: Apple explains that PyTorch can use Metal Performance Shaders (MPS) backend for GPU acceleration on Mac, with kernels optimized for Metal GPU families.

Relevanz für NALA-DNA-Med: Supports Local Lite Mode on MacBook Pro, but with a CPU fallback because not every model/op is guaranteed to behave identically.

2. PyTorch Documentation: MPS Backend

https://docs.pytorch.org/docs/stable/notes/mps.html

Original-Kernaussage / Zusammenfassung: PyTorch notes describe the MPS backend as extending the PyTorch ecosystem so scripts can set up and run operations on GPU on macOS devices.

Relevanz für NALA-DNA-Med: Source for device detection logic: torch.backends.mps.is_available(), then fallback to CPU.

Cheminformatics / Molecule Tools

1. RDKit GitHub Repository

https://github.com/rdkit/rdkit

Original-Kernaussage / Zusammenfassung: RDKit is described as a collection of cheminformatics and machine-learning software written in C++ and Python, supporting molecular operations, descriptors, fingerprints and Python wrappers.

Relevanz für NALA-DNA-Med: Future plugin dependency for SMILES validation, molecule descriptors, fingerprints and molecule preprocessing.

2. RDKit Documentation: Getting Started in Python

https://www.rdkit.org/docs/GettingStartedInPython.html

Original-Kernaussage / Zusammenfassung: RDKit getting-started page gives an overview of using RDKit functionality from Python and is framed as an introductory guide rather than a full manual.

Relevanz für NALA-DNA-Med: Good developer onboarding source for Codex and future NALA-DNA-Med plugin work.

Application Backend / API

1. FastAPI Documentation

https://fastapi.tiangolo.com/

Original-Kernaussage / Zusammenfassung: FastAPI is a modern Python web framework for building APIs with Python type hints, interactive documentation and high performance characteristics.

Relevanz für NALA-DNA-Med: Suitable framework for nala-api and biomodel-api because it is simple, typed, testable and Swagger-friendly.

Databases / Storage

1. PostgreSQL Documentation

https://www.postgresql.org/docs/

Original-Kernaussage / Zusammenfassung: PostgreSQL documentation covers the relational database system used for structured metadata, transactions, users, projects and audit records.

Relevanz für NALA-DNA-Med: Best fit for tenants/users/projects/jobs/audit_events tables.

2. MinIO Documentation

https://min.io/docs/minio/container/index.html

Original-Kernaussage / Zusammenfassung: MinIO documentation covers S3-compatible object storage in containerized deployments and related administration/deployment concepts.

Relevanz für NALA-DNA-Med: Fits upload/file vault storage for PDFs, CSVs, FASTA files and generated reports.

Queue / Jobs

1. Redis Documentation

https://redis.io/docs/latest/

Original-Kernaussage / Zusammenfassung: Redis documentation covers Redis as an in-memory data store often used for caching, queues and fast coordination tasks.

Relevanz für NALA-DNA-Med: Useful for JobForge queue: background BioModel tasks, indexing jobs, document parsing and status updates.

Frontend / App Build

1. Vite Documentation

https://vite.dev/guide/

Original-Kernaussage / Zusammenfassung: Vite documentation describes a fast frontend build tool and development server for modern web projects.

Relevanz für NALA-DNA-Med: Good default for React + TypeScript UI, fast enough for Codex-generated MVP and local iteration.

Frontend / UI Styling

1. Tailwind CSS Documentation

https://tailwindcss.com/docs

Original-Kernaussage / Zusammenfassung: Tailwind CSS documentation describes utility-first CSS classes for rapidly building custom UI designs directly in markup.

Relevanz für NALA-DNA-Med: Fits NALA Black Edition UI: gunmetal surfaces, emerald neon accents, responsive dashboard cards.

Security / Authentication

1. Authentik Documentation

https://docs.goauthentik.io/

Original-Kernaussage / Zusammenfassung: Authentik documentation covers open-source identity provider functionality such as authentication, SSO and user management.

Relevanz für NALA-DNA-Med: Possible later identity layer for multi-user access, 2FA and external users. Not needed for v0.1, but useful for committee-grade deployment.

2. Keycloak Documentation

https://www.keycloak.org/documentation

Original-Kernaussage / Zusammenfassung: Keycloak documentation covers open-source identity and access management for modern apps and services.

Relevanz für NALA-DNA-Med: Alternative to Authentik for mature SSO/realm/role based access management.

4. Quellenmatrix für Ausschuss / Arztgespräch

Bereich	Quelle	Warum wichtig
Biomedical Foundation Model / MAMMAL	GitHub: BiomedSciAI/biomed-multi-alignment	Primary technical source for later MAMMAL integration. For v0.1 the app should only include a stub service and integration notes, not auto-download large model weights.
Biomedical Foundation Model / MAMMAL	Hugging Face: ibm-research/biomed.omics.bl.sm.ma-ted-458m	Reference for model download location, model size family (~0.5B), intended input domains, and future plugin target.
Biomedical Foundation Model / MAMMAL	arXiv: MAMMAL - Molecular Aligned Multi-Modal Architecture and Language	Scientific background for the BioLab concept. Use as citation for why molecule/protein/omics workflows belong in the app, while still labelling outputs as research-support only.
Biomedical Foundation Models / IBM	IBM Research: Biomedical Foundation Models	High-level institutional context for committees/doctors: this approach is an active research direction, not a random hobby stack.
Multitenancy / Data Isolation	Qdrant Documentation: Multitenancy	Core source for RAGVault design: every vector payload must include tenant_id, user_id, project_id and document_id, and every retrieval query must enforce filters.
Multitenancy / Data Isolation	Qdrant Article: How to Implement Multitenancy and Custom Sharding	Supports the TenantShield rule: shared infrastructure is OK, but every stored vector/document must carry tenant metadata.
Multi-User Frontend / RBAC	Open WebUI Documentation: Features / Authentication & Access	Good MVP option for a ready-made multi-user chat UI before building a fully custom NALA Black Edition interface.
Multi-User Frontend / RBAC	Open WebUI Documentation: Role-Based Access Control	Reference for roles such as Admin, User, Pending, and for designing doctors/researcher/viewer/admin workflows.
Multi-User Frontend / RBAC	Open WebUI Documentation: Roles	Useful for CAVEMAN-level role mapping: Owner/Admin/Doctor/Researcher/Viewer/Guest/Pending.
LLM Serving / Model Backend	vLLM Documentation: OpenAI-Compatible Server	Supports NALA-cORe backend strategy: serve local LLMs behind an OpenAI-compatible interface for UI/agent compatibility.
LLM Serving / Model Backend	vLLM Documentation: Quickstart	Good source for why NALA can support interchangeable local/remote LLM backends with the same app interface.
macOS Runtime / Container Mode	OrbStack Docs: Docker Containers	Main source for choosing OrbStack Core Mode for Qdrant/Postgres/MinIO/Redis/BioModel workers on Mac.
macOS Runtime / Container Mode	OrbStack Quick Start	CAVEMAN install source: install OrbStack, run Docker Compose, then start NALA Core Mode.
macOS Runtime / Container Mode	OrbStack Native Files / Container, Image, Volume Files	Important UX benefit for Mike/NALA: users can inspect container files without deep Docker kung-fu.
Apple Silicon / Local Inference	Apple Developer: Accelerated PyTorch Training on Mac	Supports Local Lite Mode on MacBook Pro, but with a CPU fallback because not every model/op is guaranteed to behave identically.
Apple Silicon / Local Inference	PyTorch Documentation: MPS Backend	Source for device detection logic: torch.backends.mps.is_available(), then fallback to CPU.
Cheminformatics / Molecule Tools	RDKit GitHub Repository	Future plugin dependency for SMILES validation, molecule descriptors, fingerprints and molecule preprocessing.
Cheminformatics / Molecule Tools	RDKit Documentation: Getting Started in Python	Good developer onboarding source for Codex and future NALA-DNA-Med plugin work.

Bereich	Quelle	Warum wichtig
Application Backend / API	FastAPI Documentation	Suitable framework for nala-api and biomodel-api because it is simple, typed, testable and Swagger-friendly.
Databases / Storage	PostgreSQL Documentation	Best fit for tenants/users/projects/jobs/audit_events tables.
Databases / Storage	MinIO Documentation	Fits upload/file vault storage for PDFs, CSVs, FASTA files and generated reports.
Queue / Jobs	Redis Documentation	Useful for JobForge queue: background BioModel tasks, indexing jobs, document parsing and status updates.
Frontend / App Build	Vite Documentation	Good default for React + TypeScript UI, fast enough for Codex-generated MVP and local iteration.
Frontend / UI Styling	Tailwind CSS Documentation	Fits NALA Black Edition UI: gunmetal surfaces, emerald neon accents, responsive dashboard cards.
Security / Authentication	Authentik Documentation	Possible later identity layer for multi-user access, 2FA and external users. Not needed for v0.1, but useful for committee-grade deployment.
Security / Authentication	Keycloak Documentation	Alternative to Authentik for mature SSO/realm/role based access management.

5. CAVEMAN Kurz-Argumentation

Für Ärztinnen/Ärzte:

NALA-DNA-Med soll nicht Arzt oder Labor ersetzen. Es soll Literatur, Dokumente, Molekül-/Protein-Daten und Forschungsnotizen lokal strukturieren, mit Quellen verbinden und Hypothesen sichtbar machen. Jede Ausgabe bleibt prüfpflichtig und nicht-klinisch.

Für Ausschüsse / Finanzierung:

Das Projekt nutzt vorhandene Open-Source-Bausteine und trennt MVP von späterer Hochrisiko-Funktionalität. Datenschutz und Mandantenfähigkeit werden im Datenmodell und in der Architektur von Anfang an verankert. Dadurch kann man klein starten und kontrolliert wachsen.

Für Codex / Entwicklung:

Zuerst Repo-Struktur, Doctor/Install/Start/Backup-Scripts, API/UI, TenantShield, Stub-BioModel und Tests. Erst danach echte MAMMAL/RDKit-Integration als Plugin-Branch.

6. Offene Punkte / Nicht als bewiesen behandeln

1. Die exakte lokale Laufbarkeit von MAMMAL auf dem konkreten MacBook Pro muss getestet werden. Apple-MPS kann helfen, aber CPU-Fallback bleibt Pflicht.

2. MAMMAL/RDKit/biomedizinische Ausgaben sind Forschungs- und Screening-Hilfen, keine validierten medizinischen Entscheidungen.

3. Für echte Multi-User-Nutzung ausserhalb des Heimnetzes braucht es zusätzliche Härtung: HTTPS, 2FA, Audit, Backup/Restore, Rollenprüfung, Rate Limits und Netzwerksegmentierung.

4. Für klinische Nutzung wären Validierung, Datenschutzprüfung, regulatorische Klassifizierung und medizinische Verantwortung separat zu klären.